

What drives mineral forest soil carbon sink dynamics at national scale?

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1 Fuel for attribution analysis

Consider the national soil carbon balance F using an intuitive single-pool model

$$F = \frac{dC}{dt} = L(\text{Vol}) + I(H, U) - k(T, P) C, \quad (1)$$

where C is the soil carbon stock, L is litter input from living vegetation (trees + ground vegetation), I is litter input originating from harvest residues, and k is the decomposition rate. For simplicity, let's assume litter production depends only on forest growing stock volume (Vol), while harvest-related litter input I is function of harvest volume (H) and the fraction of harvest residues left to the site (U):

$$I = \alpha H U, \quad (2)$$

where α is a conversion factor from harvested biomass to carbon entering the soil. Further, let's set decomposition rate (k) the conventional function of temperature (T) and precipitation (P).

To explore how much of the observed (or modelled as in GHGI) sink change $\frac{\partial F}{\partial t}$ over finite time period dt can be attributed to changes in volume, harvests / residue use, and climate can be done using the total differential of Eq. 1

$$dF = dL + dI - C dk - k dC, \quad (3)$$

where:

$$dL = \frac{\partial L}{\partial \text{Vol}} d\text{Vol}, \quad (4)$$

$$dI = \frac{\partial I}{\partial H} dH + \frac{\partial I}{\partial U} dU = \alpha U dH + \alpha H dU, \quad (5)$$

and

$$dk = \frac{\partial k}{\partial T} dT + \frac{\partial k}{\partial P} dP. \quad (6)$$

20 Substituting into Eq. 3 gives

$$\begin{aligned}
 dF = & \underbrace{\frac{\partial L}{\partial \text{Vol}} d\text{Vol}}_{\text{forest growing stock}} + \underbrace{\alpha U dH}_{\text{harvest intensity}} + \underbrace{\alpha H dU}_{\text{residue management}} \\
 & - \underbrace{C \frac{\partial k}{\partial T} dT}_{\text{temperature effect}} - \underbrace{C \frac{\partial k}{\partial P} dP}_{\text{precipitation effect}} - \underbrace{k dC}_{\text{stock feedback}} .
 \end{aligned} \tag{7}$$

Eq. 7 is the decomposition of soil carbon sink variability into contributions from (i) changes in forest biomass and litter production, (ii) changes in harvest intensity, (iii) changes in harvest-residue management, (iv) climate-driven changes in decomposition, and (v) the stock feedback associated with the existing soil C stock. Note that the climate contribution enters
 25 through $-C dk$, meaning that the sensitivity of the soil C balance to climate change is *always* proportional to the size of the existing soil C stock.

The above differential for single-pool model is intuitive, but can be easily extended to N pools (e.g. Yasso -family). In a special case where $\frac{\partial k_n}{\partial T}$ & $\frac{\partial k_n}{\partial P}$ are constant among pools (as in Yasso07, 15 and new 20), eq. 7 applies when $C = \sum_{n=1}^N C_n$. One can (should) add also natural mortality as separate term in eq. 1 & 7.